

Financing wind energy projects: An extended theory of planned behavior approach to explain private households' wind energy investment intentions in Germany

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ABSTRACT

Socially responsible investment (SRI) is an investment strategy which complies with ethical, social, environmental or corporate governance criteria. Investments in renewable energies are commonly regarded as complying with SRI principles. This study aims to deepen the understanding of individuals' intentions to invest in wind energy. As an underlying framework the Theory of Planned Behavior's constructs attitude, subjective norm and perceived behavioral control were used and statistically tested with an online-survey of 592 participants in 2014. Further, this theory was extended by adding the constructs consumption profile and investor profile to the framework. With the inclusion of the two additional constructs the theoretical model turns to be very suitable as it doubled the share of explained variance of behavioral intention from 25% to 50% compared to the Theory of Planned Behavior. The analysis further shows that subjective norms, perceived behavioral control, consumption profile and investor experience have statistically significant influence on investment intentions in wind energy. However, attitudes towards wind energy investments are not a significant predictor of investment intentions. We conclude that, apart from the scientific implications of our results, our findings have practical applications for the conceptual design of wind energy investments.

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1. Introduction

1.1. Investments in wind energy

The signatory states to the United Nations Framework Convention on Climate Change (2015) agreed on the Paris Climate Agreement with binding climate targets to reduce greenhouse gas emissions and with the aim of slowing global warming and mitigating its consequences. More specifically, the states negotiated and agreed upon on the goal of "[...] holding the increase in the global average temperature to well below 2 °C above pre-industrial levels and pursuing efforts to limit the temperature increase to 1.5 °C above pre-industrial levels" [1; p. 22]. In order to reach the minimum goal of keeping global warming below 2° Celsius an "energy technology revolution" is necessary [2]. This "energy

technology revolution" requires an immense volume worth of cumulative investments in energy supply [3–5]. Those investments are commonly regarded as complying with socially responsible investment (SRI) principles. SRI is an investment strategy which complies with ethical, social, environmental or corporate governance criteria [6]. A very common example of SRI is the field of wind power, which is constantly evolving [71] and is gaining new importance as a main source for energy production due to the diffusion of renewable energies (RE). In fact, the International Energy Agency estimates that \$39 trillion worth of cumulative investments in energy supply will be required worldwide [2]. Of this amount, \$8 trillion are necessary for the RE sector, of which wind energy requires most new investments (\$3 trillion) [2].

In Germany, annual investments in wind energy increased from €2.5 billion in 2007 to €11 billion in 2014, but afterwards annual investments have fallen to one third being €3.6 billion in 2019 [7]. In addition, gross electricity production from wind energy increased from 40.507 GWh in 2007 to 80.624 GWh in 2015 and even to 125.894 GWh in 2019 contributing 42.1% to the total gross energy production in 2019 [7].

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The required investments can be provided by the public sector as well as by the private sector. The private sector is largely interested in investing in renewables, as for example the study of Curtin et al. [8] found out that 78% of the sample had at least some interest in investing in various renewables (e.g., wind onshore) in Ireland. Thus, individuals are expected to contribute a substantial share of the financial resources needed as institutional investments in combination with taxation and government expenditure cannot provide all necessary investments [9]. This is similar to the conventional capital market, investments in wind energy offer a variety of investment opportunities. For individuals' current forms of investments are closed funds and participation rights. In the case of closed funds, investors are usually entitled through a limited partnership which bundles the capital of several investors and thus finance already existing or planned wind energy projects. In the case of participation rights, individuals acquire rights to the reinsurance of the company, but have no voting rights. In addition, bonds and shares are opportunities to financially participate in wind energy projects. In the case of Germany, typical participation rights in wind energy projects provide a minimum investment term of three years and start at €1,000 of invested capital. In the field of private financing, individuals have provided an important source of finance for RE projects or more specifically for wind energy [70]. For example, private households have contributed a significant share of global climate financing in 2012 with investments of \$33 billion in RE [10]. In addition, private households invested around 16% of all private investments in renewable energies between 2013 and 2016 [11] and can therefore contribute to ambitious global climate targets [12,13].

1.2. Scientific research of individuals' investment behavior

Due to the rapid increase of global investments in SRI and more specific in wind energy on the one hand and to the high investment need to meet global climate targets, SRI has not only attracted the attention of market practitioners but also the interest of academics from various disciplines who agree about the necessity of additional knowledge in order to conclusively explain individuals' SRI investment behavior [14]. Thus, studies in the field of behavioral finance have become more and more popular. Behavioral finance is a special discipline of behavioral economics, which deals with supposedly irrational behavior on financial and capital markets. Behavioral finance is a special discipline of behavioral economics, which deals with supposedly irrational behavior on financial and capital markets. This discipline has provided insight into individuals' investment behavior, particularly the psychological processes that influence individuals' investment decisions [15]. The main focus of behavioral finance theories is the analysis of behavioral biases or the main variables affecting an individual's investment decisions.

A huge number of studies exist in the field of financial investment behavior in SRI. Most of these studies deal with institutional or market professionals' behavior on stock market [16–18]. However, there is only a spare number of studies examining individuals' intentions to invest in SRI where individuals are described as non-market professionals. Thus, existing studies either deal with SRI in general without focusing on a specific aspect of the wide field of SRI [19], or use typical behavioral decision theory without extending the established framework to gain information about further influencing factors [20].

So far, there have been published only a few studies investigating individuals' intention to invest in wind energy. Gamel et al. [21] showed with a conjoint analysis that pro-environmental attitudes were positively linked to hypothetically wind energy investments. Furthermore, Gutsche and Ziegler [22], found out that warm glow in the direction of sustainable investments and

environmental awareness increase willingness to choose sustainable funds at the sacrifice of higher returns from other investment options. However, the study by Bauer and Menrad [13] showed that neither environmental self-identity [23] nor personal norms concerning individual contributions to the global climate targets of the Paris climate agreement increased the likelihood of investment intentions in wind energy.

Therefore, literature call for new studies in order to promote a better understanding of the influencing factors that lead to individuals' investment intention in wind energy [24,25]. Further, Glac [24] argues that scientific literature provides “little knowledge about the individual and contextual variables that predict the investor's decision [...]. Thus, the question of “why do some investors practice SRI and others don't?” is still largely unanswered.”

Therefore, the current manuscript aims to partly fill this research gap by contributing to the scientific understanding why private investors intend to invest in wind energy projects. Consequently, the first research question is as follows:

- (1) What are the main factors influencing an individual's intention to invest in wind energy?

Moreover, it is argued that scientific literature provides “only a scantily developed theoretical basis for understanding SRI” [24]. Therefore, this manuscript contributes to the development of the theoretical basis for understanding individuals' investment intentions in SRI by using the Theory of Planned Behavior (TPB) [26] in order to examine the influencing factors of individuals' investment intention in wind energy. With respect to this methodical approach, the following research question arises:

- (2) Is the TPB a suitable model for predicting individual's investment intention in wind energy?

Further, this manuscript contributes to scientific knowledge by validating recent studies [27,28] on the influencing effect of TPB's variables on individuals' investment intentions. Ajzen and Fishbein [29] recommend incorporating additional factors into the TPB, since behavioral intention may also be influenced by other factors. Thus, this manuscript aims to improve the TPB to be more applicable for investment decisions in wind energy by extending the TPB by including two additional attributes to the theoretical model, namely *consumption profile* and *investor experience*. In order to validate the advantage of the extended theoretical model the following research question is included:

- (3) Can the inclusion of additional constructs improve the predictive power of the basic theoretical model of TPB?

The reminder of this paper is as following: In section 2 the theoretical framework of this study is described followed by details of data collection and analysis in section 3. Section 4 provides the statistical results of the model testing and in section 5 we discuss our results and the implications of this paper.

2. Theoretical model

An understanding of retail investors' choices and of the influencing factors which underpin these investment decisions is required. It has been shown that psychological and situational factors as well as individual attitudes can be important factors for predicting investment intentions (e.g., [21,22]). However, a theoretical framework which consistently identifies the determinants of intentions to invest in wind energy is needed to further promote the expansion of this technology.

The TPB [26] provides a theoretical framework for systematically determining the factors influencing voluntary decisions. The TPB is an extension of the earlier Theory of Reasoned Action (TRA) [29], which assumes that most individual's behavioral intention can be seen as a function of their *attitude* towards the intended behavior and their *subjective norm* [29]. The TRA presupposes that most behavior is under volitional control, i.e., that each individual can decide independently whether he or she performs a certain behavior or not. However, the realization of some behaviors will be restricted by a lack of opportunities, skills or resources [30]. For this reason, the TPB expands the TRA with a third variable, perceived behavioral control. Perceived behavioral control refers to an individual's perception of his or her ability to perform the behavior in question.

The TPB has been applied to various research areas relating to environmental behavior [31], energy saving [32] or recycling behavior [33]. TPB has also proven to be very successful for investment decisions dealing with e.g., the stock market [16], mutual funds [34] or SRI [19]. Thus, TPB is a suitable framework for analyzing investment decisions in wind energy. East [27] also points out that “TPB should apply [...] to financial investment and other saving decisions”.

Although many studies have demonstrated the suitability of the TPB for the investigation of individuals' decision-making, further factors may also influence these choices, especially in case of complex behaviors like investments in wind energy [21]. Consequently, the theoretical framework for identifying the determinants of investment behavior may consider a bundle of factors.

Ajzen and Fishbein [29] also recognize that behavior may also be influenced by additional factors such as personality, past experience or demographics. Therefore, the authors recommend incorporating additional explaining factors into the TPB, provided that these variables contribute to a more powerful explanation of the behavior examined. Against this background, the TPB is extended in this study by incorporating the variables *consumption profile* and *investor experience* into the model. The proposed model (extended TPB) is shown in Fig. 1.

2.1. Attitude

Ajzen and Fishbein [29] define *attitude* as the outcome of a general evaluation of a behavior. It depends on the personal rating of the consequences of this behavior and the probability of its

occurrence. Previous studies indicate that positive environmental attitudes [21] influence the intention to invest in wind energy. Furthermore, literature shows that an individual's financial attitude has a significant effect on his/her intention to invest in renewable energies [20] or more generally in SRI [19]. Therefore, the first hypothesis is postulated:

H1. Attitudes towards green investments positively influence an individual's intention to invest in wind energy.

2.2. Subjective norm

Ajzen and Fishbein [29] argue that the social pressure an individual perceives has a significant effect on his or her behavioral intentions. The *subjective norm* refers to this perceived social pressure to perform or not to perform the behavior. It consists of an injunctive and a descriptive norm. The injunctive norm refers to the perception of what is approved or disapproved of by peers, whereas the descriptive norm is the perceived peer behavior. In the field of investment behavior, *subjective norm* has been found to be a reliable predictor for behavioral intention [19,20,28]. Consequently, the second hypothesis is postulated:

H2. Subjective norms influence an individual's intention to invest in wind energy.

2.3. Perceived behavioral control (PBC)

Even if individuals have an environmentally sustainable *consumption profile*, this does not necessarily lead to actual investment behavior. They may easily be prevented from acting by the absence of suitable resources, skills or opportunities. Consistent with this approach, various studies have shown that investment behavior is likely to be influenced by the perception of constraining factors [19,20].

The first constraining factor used in our study is an individual's financial resources as this is a limiting factor for any investment. Several studies have shown the limiting influence of income [35] and access to financial resources [36]. The second factor measuring *PBC* refers to the perception of investment opportunities. Without knowing the existence of an investment opportunity, the intention to invest in wind energy may be restricted. With respect to these findings, the third hypothesis is postulated:

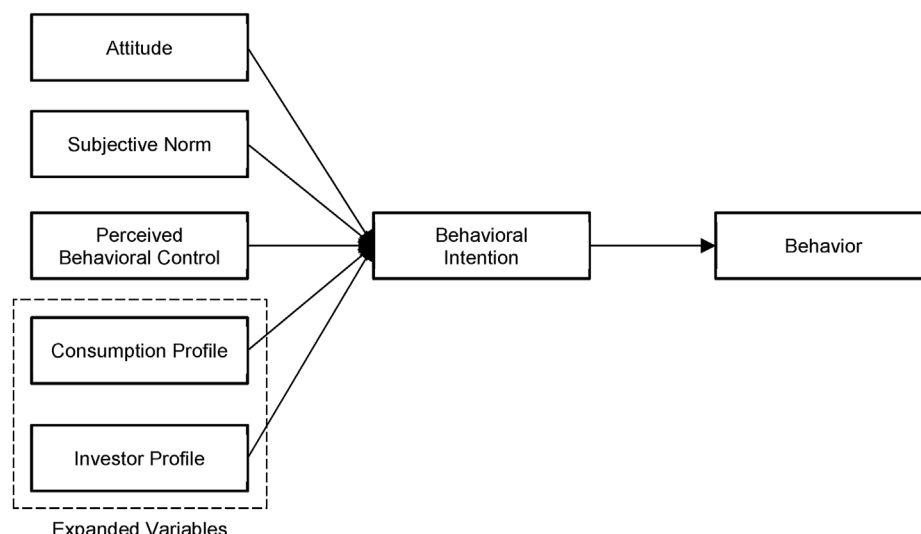


Fig. 1. Theory of Planned Behavior by Ajzen [26] with extended variables.

H3. Perceived behavioral control influences an individual's intention to invest in wind energy.

2.4. Consumption profile

Consumption profile relates to an individual's environmentally sustainable consumption [37]. It is assumed that people who consider the environmental impact of their purchase and consumption behaviors have a tendency to invest in environmentally friendly technologies such as renewable energy. Various studies in the field of SRI have identified the influencing effect of environmental attitudes on an individual's tendency to invest as well as on actual investment behavior [38] [20]. Thus, the fourth hypothesis is postulated:

H4. Green consumption profile positively influences an individual's intention to invest in wind energy.

2.5. Investor experience

Ajzen [26] argues that experience with a behavior affects an individual's *attitudes*, *subjective norms* and *perceived behavioral control* and is therefore not an independent variable of the TPB-model. Previous studies have shown that past experience does influence both investment intention and/or investment behavior [39]. In addition, an individual's product knowledge has an influencing effect on investment behavior, with several studies pointing out the significant direct impact on an individual's decision to invest in shares [17] as well as in environmental technologies [40]. Our study incorporates the variables past experience in renewable energy investments and knowledge in real value investment as predictor variables for the latent variable *investor experience*. With respect to previous studies identifying past experience as an influencing factor, the following hypothesis is postulated:

H5. Investor experience influences an individual's intention to invest in wind energy.

3. Material and methods

3.1. Measurement scales

The scales used to measure the latent variables were based on previous literature and applications of the TPB as well as information obtained from 10 expert interviews with financial service providers, consultants for renewable energies and wind energy project developers. Ajzen [26] recommends a seven-point rating scales to measure the components of the TPB. In our study, the latent variables *attitude*, *subjective norm*, *consumption profile* and *behavioral intention* were measured with the help of bipolar rating scales. We used five-point Likert scales ranging from 1 (totally disagree) to 5 (totally agree) to operationalize these variables as this offered enough answer alternatives without overwhelming the interviewees [41]. The statements for measuring the latent variables are shown in Table 1.

The first latent variable was the individual's *attitude* towards wind energy investments. For this variable, a set of six statements (Table 1) was developed according to information provided in the expert interviews.

The latent variable *subjective norm* refers to the perceived social pressure to perform or not to perform the behavior in question [29]. It was operationalized by a set of six statements (Table 1) adopted from previous studies [28,33] and adapted to the scope of this study. Three of the statements regarded the perceptions of related people about wind energy and the remaining three statements regarded individuals' perceptions about SRI.

The latent variable *PBC* can be operationalized by directly asking respondents about the extent of control they feel they have about their intended behavior. Another way of operationalizing this variable is to use an indirect measure, which assesses the constraining factors of the behavior, an approach which is recommended by Parker et al. [42]. Our study uses the latter approach. Two questions were included which addressed *PBC* in the model. First, individuals' financial resources were elicited by asking participants to indicate the amount of their assets. The measurement scale was adopted from the German Federal Statistical Office [43]) and included five levels ranging from less than 2500 € to more than 500,000 €. The second item measured the extent to which an individual notices investment opportunities, a factor which can also have a constraining effect on the intention to invest. Respondents were asked whether they had noticed a concrete opportunity to invest in wind energy in the past (Table 1). The scale of this variable was nominal with answers only including "yes" or "no".

In order to measure the latent variable *consumption profile*, a set of six statements defining the Green Consumer Value according to Bearden et al. [37] was used (Table 1).

The latent variable *investor experience* can be operationalized by an individual's past experience with similar investments and by the knowledge about those investments. The first question elicited the presence of previous investments in renewable energies. Answers were limited to "yes" or "no". In order to investigate the individual's product knowledge, respondents were asked to assess their current state of knowledge about real value investments. The measurement scale and the wording of statements were adopted from a questionnaire on investment typology from the German investment company [44]. Answers were provided on a four-point Likert scale with values ranging from 1 (no experience) to 4 (a lot of experience). The concrete wording used for the two statements defining *investor experience* is shown in Table 1.

In order to measure the latent variable *behavioral intention*, Ajzen [45] recommends the use of a statement describing a very specific example of an action. The intention to invest in wind energy was elicited using two statements (Table 1) describing a specific investment behavior of 3000 € or 10,000 € in the coming 12 months.

3.2. Measurement model

In order to test the hypotheses as well as the explanatory power of TPB in predicting individuals' intention to invest in wind energy, structural equation modeling (SEM) analysis was used. SEM has mainly been used in the economic and social sciences since the early 1970s. Previous studies successfully adopted SEM for the evaluation of cause-effect relationships in diverse areas such as the success in corporate governance [46]; image research in marketing [47]); and simulation models in hydrology research [48,49]. The application of SEM is also established in the area of socially responsible consumer behavior with various studies demonstrating the applicability of SEM to energy saving behavior [32], recycling behavior [50], adoption of photovoltaics [20], and SRI behavior [19].

SEM enables relationships among multiple independent and dependent constructs to be modeled simultaneously [51,52] in a single, systematic and comprehensive analysis. The combined calculation of the measurement and the structural model enables "measurement errors of the observed variables to be analyzed as an integral part of the model, and factor analysis to be combined in one operation with the hypotheses testing" [52]). In this way, SEM takes the measurement error into account while estimating the modeled path coefficients [51].

To assess the strength of the cause-effect relationships between latent variables, the theoretical constructs have to be measured by

Table 1
Statements for measuring the latent variables.

Designation	Statement
Attitude	
att_1	When investing, it is important to me to invest in ethically correct companies.
att_2	Investments should be ethically and morally acceptable.
att_3	When investing, I pay attention to the environmental friendliness of companies and projects.
att_4	When investing, I would renounce higher returns in favor of sustainability aspects.
att_5	Investments should be scrutinized for their environmental compatibility.
att_6	When investing in a company, it should be ecologically harmless.
Subjective norm	
norm_1	Most of my friends think positively about wind energy.
norm_2	Most people who are important to me would approve of the construction of a wind turbine.
norm_3	My family members are disturbed by the view of a wind turbine.
norm_4	Most of my friends would invest in sustainable investments.
norm_5	Most of the people who are important to me think that sustainable investments make sense.
norm_6	My family members consider it important that some of their assets will be invested in sustainable projects.
Perceived Behavioral Control (PBC)	
pbcc_1	What is the value of your assets?
pbcc_2	Have you noticed investment offers in wind energy within the previous 24 months?
Consumption profile	
cons_1	It is important to me that the products I use do not harm the environment.
cons_2	I consider the potential environmental impact of my actions when making many of my decisions.
cons_3	My purchase habits are affected by my concern for our environment.
cons_4	I am concerned about wasting the resources of our planet.
cons_5	I would describe myself as environmentally responsible.
cons_6	I am willing to be inconvenienced in order to take actions that are more environmentally friendly.
Investor experience	
invest_1	Have you ever invested in renewable energies before?
invest_2	What knowledge or experience do you have in the field of real assets (e.g., shares, open equity funds, precious metals)?
Behavioral intention	
inten_1	I intend to invest approximately 3000 € in wind energy within the next 12 months.
Inten_2	I intend to invest approximately 10,000 € in wind energy within the next 12 months.

observable indicators. The evaluation of these latent variables is possible through the use of reflective and/or formative measurement models. According to Diamantopoulos et al. [53], it has been demonstrated that SEM's measurement models are often operationalized incorrectly due to a knowledge gap of causal effect relationships. To ensure a proper operationalization of the measurement models, the latent variables used in our study were defined as reflective and formative constructs regarding their cause-effect relationship. Furthermore, path-modelling software which can handle both types of constructs was used.

In the case of reflective constructs, the values of the observable variables are caused by the latent variable [54]. This is accompanied by the assumption that changes in the unobservable variable cause changes in all observed indicators equally. In addition, the indicators should have the same parent content or at least a common theme [55]. Since these conditions apply for the constructs *attitude*, *consumption profile* and *behavioral intention*, they were defined as reflective.

In contrast to the reflective model, the essential feature of a formative measurement model is an inversed direction of influence. In the case of formative constructs, the observable variables cause the latent variable [54]. Thus, there is no need for the indicators of a formative construct to measure the same content or to have a common theme [54]. Since the indicators of the construct's *subjective norm*, *PBC* and *investor experience* causally influence the related latent variables and do not measure the same content, they were defined as formative. Appendix A shows the proposed structural model with the inclusion of the reflective (*attitude*, *consumption profile*, *behavioral intention*) and formative (*subjective norm*, *PBC*, *investor experience*) constructs.

3.3. Data collection

Before the online survey was conducted from June to July 2014, an extensive pretest was carried out in order to test the comprehensibility of the questionnaire. The target population was made up of German individuals aged 18 or above. The online questionnaire was set up by us and then a market research company was subcontracted to recruit the respondents. Participants were randomly selected by the company and invited to participate via email. The email invitation included no indication of the survey topic but a web link to the introduction page of this survey. From a total of 18,736 invitations, 11,726 individuals accepted the invitation. After excluding incomplete responses and respondents with predictable answer patterns, a sample of 2024 individuals remained. As the latent variable *PBC* is measured *inter alia* by the individual's financial resources, our study is based on a subsample of participants who voluntarily provided information about their assets. Finally, 592 individuals (final sample) were deemed to be suitable for our analysis. The socio-demographic characteristics of the final sample in comparison to the population are shown in Table 2. The study sample contains a small over-representation of male (57%) respondents. The age distribution reveals a predominance of individuals in the range of 50–69 years (53%). Respondents aged up to 30 years or above 70 years only represent a small share of the respondents. Most of the respondents have a university or college degree (39%), in contrast to the population, in which most people have a secondary modern school degree. The sample's distribution of net household income per month agrees fairly well with the sampling frame.

Table 2
Socio-demographic characteristics of the sample.

Variable	Sample	Population (Sampling frame)
Gender		
Male	0.57	0.49 ^a
Female	0.43	0.51 ^a
Age		
18 to under 30 years	0.08	0.17 ^a
30 to under 40 years	0.15	0.14 ^a
40 to under 50 years	0.18	0.20 ^a
50 to under 60 years	0.30	0.17 ^a
60 to under 70 years	0.23	0.14 ^a
70 years or older	0.06	0.18 ^a
Education		
Without school degree	0.00	0.00 ^a
Secondary modern school degree	0.13	0.42 ^a
High school degree	0.29	0.30 ^a
Academic high school degree	0.19	0.13 ^a
University or college degree	0.39	0.15 ^a
Net household income per month		
Under 900 €	0.03	0.04 ^b
900 to under 1.300 €	0.06	0.06 ^b
1.300 to under 1.500 €	0.05	0.05 ^b
1.500 to under 2.000 €	0.14	0.12 ^b
2.000 to under 2.600 €	0.17	0.17 ^b
2.600 to under 3.600 €	0.25	0.24 ^b
3.600 to under 5.000 €	0.20	0.21 ^b
5.000 € and more	0.10	0.11 ^b

^a [6].

^b [43].

4. Results

In order to test the hypotheses and to assess the validity of the scales, SEM software (SmartPLS), which makes use of partial least square (PLS) path modelling, was applied. PLS is a component-based modelling technique for estimation purposes which has minimal restrictions on measurement scales, sample size and residual distributions [56]. Furthermore, PLS can handle reflective and formative factors [52]. Thus, PLS was chosen to allow the integration of the formative factors in the equation model.

In order to ensure a systematic evaluation of the structural model, the reflective and formative measurement models were examined in a first step. Following this, the relationships of the latent variables in the inner structural model were analyzed. For each evaluation, the validation guidelines provided by various studies [53,56] are used for orientation. To determine the significance of the paths among the constructs, the “Bootstrapping-procedure” was used. This procedure estimates the mean values and standard errors based on the estimation of 500 subsamples. Thus, significances for item loadings and path coefficients β are derived with the use of t-statistics. Further, Appendix B provides descriptive statistics of the variables included within the structural model.

4.1. Model accuracy

4.1.1. Reflective measurement models

Attitudes towards behavior, the individual's consumption profiles and the behavioral intention to invest in wind energy were operationalized by reflective indicators. Therefore, the measurement models were tested for indicator reliability, internal consistency reliability, convergent validity, and discriminant validity. The results are based on a confirmatory factor analysis (CFA) procedure in PLS [57]. When testing for indicator reliability, at least half of the variance of an indicator should be explained by the associated

construct. This corresponds to a minimum loading of 0.7 and an indicator reliability value of 0.4. Table 3 reveals that all indicators measuring the reflective latent variables had individual indicator reliability values larger than the minimum level of 0.4 and close to the preferred level of 0.7 and therefore indicator reliability can be confirmed.

Constructs can be regarded as reliable indicators if they correlate in a positive and strong manner. Usually, Cronbach's α [58] is used to measure construct reliability in social science research but it tends to underestimate the internal consistency in PLS-SEM [59]. Other studies have suggested that using composite reliability with values > 0.6 is acceptable instead [59]. The composite reliability values of the three reflective constructs are all larger than 0.6 (Table 3) and can be considered to be reliable.

To check for convergent validity the variable's Average Variance Extracted (AVE) is calculated. AVE is a more stringent measure of quality than the construct reliability and its values should be greater than the acceptable threshold of 0.5. As it can be seen in Table 3, all reflective construct's values meet this criterion and therefore convergent validity is confirmed.

Discriminant validity demonstrates the extent to which indicators of one construct differ to the indicators of another construct. According to the Fornell-Larcker criterion [60], the AVEs square root in each latent variable should be larger than all other correlation values among the latent variables. Therefore, a matrix of loadings and cross-loadings (Appendix C) was created which confirms discriminant validity for the formative operationalized variables as it reveals that all indicators load higher on their assigned constructs than on any other construct. As such, all reflective constructs can be considered as reliable and valid.

4.1.2. Formative measurement models

The latent variables *subjective norm*, *PBC* and *investor experience* were operationalized by formative constructs. This implies that the indicators cause the latent variables. Therefore, the manifest variables must be independent and must explain the latent variables as much as possible. Due to the reversal of causality, the evaluation of formative measurement models requires different criteria to those of reflective models. Outer model weight and significance, convergent validity and collinearity among the indicators used to evaluate the models' quality.

The outer model weights can be interpreted as regression coefficients. Values close to 1 and -1 indicate a strong relationship between the indicators and constructs, whereas values close to zero indicate a weak relationship. The outer model weights were tested for significance using the “bootstrapping-procedure”. The t-test was used to check whether the outer weights were significant at the 5%-level. The mean values and significance levels are presented in Table 4. They reveal that all indicators except *norm_4* and *norm_5* have a relatively strong correlation with the corresponding constructs.

Formative measurement models are based on the same principle as regression analysis. As such, the model estimation is less reliable the more standard errors and beta-coefficients rise due to multicollinearity. The Variance Inflation Factor (VIF) indicates the factor by which the variance of parameter estimates is inflated due to multicollinearity. According to Diamantopoulos et al. [53], VIF values of 1 are indicative of no multicollinearity, whereas values > 10 are critical. However, Ringle and Spreen [52] argue that VIF values < 3 should be more conservatively interpreted. All indicators of the formative measurement models fulfilled the requirements of VIF < 3 , demonstrating the non-existence of multicollinearity. In summary, all accuracy criteria of formative measurement models are fulfilled in our study as shown in Table 4.

Table 3
Accuracy measures of reflective latent variables.

Latent Variable	Indicators	Loadings	Indicator Reliability	Composite Reliability	AVE
Attitude	atti_1	.861	.741	.937	.715
	atti_2	.852	.726		
	atti_3	.870	.757		
	atti_4	.741	.549		
	atti_5	.870	.757		
	atti_6	.871	.759		
Consumption profile	cons_1	.825	.681	.926	.678
	cons_2	.869	.755		
	cons_3	.853	.728		
	cons_4	.753	.567		
	cons_5	.841	.707		
	cons_6	.793	.629		
Behavioral intention	inten_1	.967	.935	.964	.930
	inten_2	.962	.925		

Table 4
Accuracy measures of formative latent variables.

Latent Variable	Indicators	VIF (<3)	t-value (≥ 1.96)	Outer weight (≥ 0.1)
Subjective norm	norm_1	2.403	2.099	.293
	norm_2	2.594	1.544	.205
	norm_3	1.117	7.711	-.640
	norm_4	2.614	1.052	-.147
	norm_5	2.668	0.807	.114
	norm_6	1.995	4.982	.603
PBC	pbc_1	1.006	5.662	.536
	pbc_2	1.006	12.108	.805
Investor experience	invest_1	1.029	10.091	.521
	invest_2	1.029	19.568	.771

4.1.3. Inner path model

In order to examine the explanatory power of the TPB or rather the extended TPB in predicting individuals' intention to invest in wind energy, the percentage of variance explained by the model (R^2) was considered. R^2 can have values between 0 and 1 and should be correspondingly high, depending on the specific research question. According to Chin [56], values of 0.19, 0.33 or 0.67 can be described as “weak”, “moderate” or “substantial”.

The coefficient of determination, R^2 , is presented in Table 5 for both models. The TPBs latent variables ATTI, NORM and SITU only explained 25% of variance, which can be considered weak. It must be pointed out that in a model with only a few variables, R^2 cannot be too high because many other determinants may have been disregarded [61]. Nonetheless, the inclusion of the latent variables CONS and INVEST increases the extended TPBs R^2 to 50%, which can be described as moderate.

Another indicator to test for forecast quality is Cohen's effect size f^2 [62]), which determines the degree to which the

phenomenon is present in the population [56]. Ringle and Spreen [52] interpret f^2 as the degree of substantial influence from exogenous latent variables on the endogenous latent variable. f^2 values of 0.02, 0.15 or 0.35 are described as having low, moderate or high influence respectively on the endogenous latent variable. Within the TPB, all latent exogenous variables influence *behavioral intention* on a low level (Table 5). In the extended TPB, it can be seen that *investor experience* (0.485) has a high influence on individuals' intention to invest in wind energy, whereas the remaining variables have low influence on this intention.

The predictive capabilities of the structural models were tested using the cross-validated redundancy measure Q^2 . This measure provides information about how well the empirically collected data can be reconstructed by the model developed and parameter estimations without losing degrees of freedom. If the value for Q^2 is greater than zero, it is assumed that the model has forecast relevance. The Q^2 value of the TPB model is 0.217 and therefore forecast relevance can be accepted. However, the extended TPB has a Q^2

Table 5
Path coefficients β , t-statistics and f^2 of the inner path model.

	Theory of Planned Behavior ^a			Extended Theory of Planned Behavior ^b		
	β	t-statistic	f^2	β	t-statistic	f^2
Attitude	.054	1.329	.003	.002	0.028	.000
Subjective Norm	.343***	8.630	.126	.229***	6.625	.078
Perceived Behavioral Control	.275***	7.507	.098	.111**	3.042	.021
Consumption Profile				.110*	2.074	.010
Investor experience				.525***	15.401	.485

*p < .05, **p < .01, ***p < .001.

^a Adj. R^2 0.247.

^b Adj. R^2 0.499.

value of 0.452 and therefore provides much stronger predictive capabilities.

4.2. Verification of hypotheses

In order to test the hypotheses, the inner model's path coefficients β have to be estimated. These coefficients are presented in Table 5. According to Ringle and Spreen [52], when latent exogenous variable's path coefficients are close to 0, this indicates a weak effect, whereas values close to 1 or -1 indicate a strong effect on the latent endogenous variable. As there are no parametric tests for significance of the path coefficients due to a lack of distributional assumptions, the "Bootstrapping-procedure" was used for this purpose. The t-statistics show that the path coefficients of both models, the TPB and the extended TPB, were statistically significant at a 95% level with the exception of *attitude*.

With respect to the hypotheses, the following statements can be made considering the inner model's path coefficients β and the corresponding significance. The *attitude* towards wind energy investments has no significant influence on the *behavioral intention* and therefore hypothesis H₁ ($\beta = 0.002$; $p > .05$) is not supported. The influence of *subjective norm* is significant and therefore hypothesis H₂ ($\beta = 0.225$; $p < .001$) is supported. Hypothesis H₃ ($\beta = 0.110$; $p < .01$) is supported as *PBC* has a significant influencing effect on the intention to invest in wind energy. An individual's *consumption profile* influences the investment intention, thus supporting hypothesis H₄ ($\beta = 0.112$; $p < .05$). Finally, the influence of an individual's experience on the *behavioral intention* is statistically significant and therefore hypothesis H₅ ($\beta = 0.525$; $p < .001$) is supported.

5. Discussion and conclusion

Due to constraints which prevented the observation of individuals' actual investment behavior, this paper only investigates individuals' intentions to invest in wind energy and not their actual behavior. Nonetheless, analyzing intentions is valuable because the execution of behavior is strongly influenced by intention [26]. To evaluate individuals' investment intentions, the well-established TPB was extended by the construct's *consumption profile* and *investor experience*. This approach is in accordance with Ajzen and Fishbein [29], who recommend five to nine attributes to explain intended behavior. However, the inclusion of additional variables to the TPB is only accepted when these variables contribute significantly to the explanation of the model [26], which is the case in our study.

5.1. Explanatory power of TPB and extended TPB

Consistent with previous research [19,32], *subjective norm* and *PBC* had significant influence on *behavioral intention*. In conjunction with *attitudes*, they made up the TPB, which explains 25% of the variance of *behavioral intention* in our study. The explained variance of the TPB in our study seems to perform similarly to that in the study of Tonglet et al. [63], whose TPB revealed an explanatory power of 26% for predicting an individual's intention to recycle household waste. Nevertheless, the TPB without the proposed extensions does not seem to predict an individual's investment intentions in wind energy, which is in contrast to East [27], who points out the suitability of the TPB for financial investment and savings decisions.

However, through the inclusion of *consumption profile* and *investor experience*, both of which have a significant effect on the *behavioral intention*, the extended TPB in our study explained 50% of the variance of *behavioral intention* in wind energy. Thus, the

explanatory power of the extended TPB is consistent with the results of previous studies [19,20] that investigated individuals' investment intentions. The gain in explanatory power is also consistent with the results of Chen's [32] study of individuals' intentions to save energy.

5.2. Predictors of investment intention

In contrast to previous studies [32,64], *attitudes* had no significant influence on *behavioral intention*. This indicates that an individual's attitude towards wind energy investments is not a good predictor of their intention to invest in wind energy. Furthermore, it is striking that the path coefficient β of the variable *attitudes* decreases to almost zero with the inclusion of *consumption profile* and *investor experience* in the model. This could be explained by the operationalization of the variable *attitude*. Ajzen [26] argues that attitude is the outcome of a general evaluation of a behavior, whereas Davies et al. [65] separates attitudes into two components: affective, which defines feelings about behavioral intention; and cognitive, which defines the personal rating of the consequences of such behavior and its probability of occurrence. In our study, *attitude* was operationalized as feelings about investments in wind energy without considering the possible consequences.

The estimation of path-coefficients β revealed that the variable *subjective norm* has a significant influencing effect on an individual's *behavioral intention* to invest in wind energy in both the TPB and the extended TPB models. Within these models, the effect size is in line with the results of comparable studies [19,20] but significantly greater than the results of Chen [32], suggesting that *subjective norm* is a better predictor for direct SRI investments (like e.g., wind energy) than for indirect SRI investments like energy-saving behavior.

The variable *PBC* was operationalized using an indirect item that assessed the constraining factors of the behavior instead of typically measuring how easy or difficult it is to perform the intended behavior. The influencing effect of *PBC* on the *behavioral intention* performs at a relatively low level in our model. This is in line with the results of Adam and Shauki [19], who, in contrast to our study, operationalized *PBC* with two statements that measured how easy or difficult it is for an individual to invest in SRI.

The results further revealed that individuals' *green consumption profile* performs a significant effect on the intention to invest in wind energy. Thus, this finding is in line with the study of Nilsson [14] who confirmed that "*pro-social attitudes regarding the issues addressed in SRI will impact consumer behavior for socially responsible investment in a positive manner*." Further, the measurement scale used by Nilsson [14] to operationalize pro-social attitudes is comparable to this study's variable *consumption profile*. Our finding is further in line with Hartmann and Apaolaza-Ibáñez [66] who argued that individuals' environmental concern performs both, a direct and an indirect effect on the behavioral investment intention towards green energy brands. Additionally, the results are supported by Follows and Jobber [67] who argue that there is a positive relationship from an individuals' attitude towards the environmental consequences to his or her behavioral intention to invest a certain amount of money. Our finding is further indirectly supported by the result of Laroche et al. [68] who state that "*consumers who consider environmental issues when making a purchase are more likely to spend more for green products*". In line with this, Getzner and Grabner-Kräuter [69] assume that "*consumers who buy green consumer products based on certain environmental attitudes would typically also consider these attitudes to distinguish between more or less preferable investments*". Thus, our results suggest that general scales like the "Green Consumer Value" [37], which are represented by the variable *consumption profile*, may perform a greater

influencing effect on an individual's investment intention than the specific *attitudes* towards the intended behavior, at least in the context of investment intentions in wind energy.

Moreover, *investor experience* was the best predictor for an individual's intention to invest in wind energy, with the influencing effect being strong and significant. This result is consistent with existing literature [17] and supports the presumption that general factors may outweigh more specific factors, at least for wind energy investment intentions.

However, it must be emphasized that one of the two items used to operationalize the variable *investor experience* was not measured objectively. Participants had to assess their own knowledge in real value investment on a 4-point Likert scale with values ranging from 1 (no experience) to 4 (a lot of experience), which is a subjective measure as the reference level for what is perceived as “no experience” or “a lot of experience” varies from person to person.

5.3. Limitations of the current study

To recruit respondents, we subcontracted a market research company which randomly invited their panel users via email to participate in our study. Since the panel members could decide whether to accept the invitation, self-selection cannot be excluded in our study. Additionally, a further selection process was made and only participants who voluntarily gave information about their assets were included. As such, the sample cannot be interpreted as having been completely randomly selected.

A further limitation which should be noted is the fact that our sample consists exclusively of German citizens. Since Germany has a global pioneering role in the expansion of renewable energies, conclusions about the influencing factors of the behavioral intention to invest in wind energy cannot be drawn about individuals living outside Germany.

In contrast to Ajzen [45], who recommends formulating five to six items to assess each of the theory's major constructs, the latent variables *PBC* and *investor experience* were operationalized by only two items. However, this approach is in line with previous studies that have shown that the operationalization of TPB variables is still applicable with fewer items [19,20].

5.4. Implications and future research

The TPB has proven to be very successful for investment decisions dealing with e.g. stock market [16], mutual funds [34] or SRI [19]. Nevertheless, the original TPB as suggested by Ajzen [26] cannot not provide satisfactory explanatory power for predicting individuals' intentions to invest in wind energy. In contrast, our proposed extended TPB, which includes the additional variables *consumption profile* and *investor experience*, appears to be a good framework for predicting individuals' investment intentions in wind energy. However, future research should address our approach by investigating variables which explain more of the variance of individuals' behavioral intention to invest in wind energy and which determines the predictive power of individuals' intended behavior on their actual investment behavior.

Moreover, our sample consists exclusively of German citizens and therefore no direct conclusions can be drawn about behavioral intentions of individuals living outside Germany. Nevertheless, the results of this study could be applied for the conception of wind energy investments in other countries. Additionally, the proposed theoretical model needs to be further considered in order to verify the applicability of the extended TPB on individual's behavioral intentions to invest in other types of renewable energies both inside and outside of Germany.

Despite various opportunities to invest for example in wind

energy funds [13], wind farms, or companies that manufacture wind turbines, private households, especially in urban areas, may not have the opportunity to install a small-scale wind turbine on their own property. In comparison, solar photovoltaics can be installed in urban areas and by a higher number of households in rural areas. Future research could therefore investigate whether the ability to install a small-scale wind turbine affects investment intentions and could compare investment intentions in solar photovoltaic and wind energy, considering the potential for realization. The focus of this manuscript was not on a specific wind energy investment opportunity.

As our study investigates individuals' intention to invest in wind energy, not their actual behavior, further research is required to investigate how well *behavioral intention* predicts actual investment behavior and if *PBC* has an influencing effect on behavior as stated by the TPB.

CRedit authorship contribution statement

Johannes Gamel: Conceptualization, Data curation, Formal analysis, Writing the first draft. **Andreas Bauer:** Writing original draft, Writing review and editing. **Thomas Decker:** Project administration, Funding acquisition, Writing original draft, Support. **Klaus Menrad:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.renene.2021.09.108>.

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